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Numerical Analysis Assignment 3.

1. (a) $2x^4-5x^3+3x^2+x-1=0$ $r=-\frac{1}{2}$ $r=1$

Solution: $f(-\frac{1}{2})=0$

$f(1)=0$

$f'(x)=8x^3-15x^2+6x+1$ $f''(x)=24x^2-30x+6$

$f'(-\frac{1}{2})=-\frac{19}{4}-3+1=-\frac{27}{4}\neq 0$

$f'(-1)=-28\neq 0$

so quadratic convergence.

$\lim_{i\to\infty}\frac{e_{i+1}}{e_i^2}=M=\left|\frac{f''(r)}{2f'(r)}\right|$

for $r=-\frac{1}{2}$ $M=\left|\frac{27}{-\frac{27}{2}}\right|=2$ $e_{i+1}\approx 2e_i^2$

for $r=1$ $M=0$ superlinearly.

(b) $x^3-x^2-5x-3=0$; $r=-1$ $r=3$

$f(-1)=0$ $f(3)=0$

$f'(x)=3x^2-2x-5$ $f''(x)=6x-2$

$f'(-1)=0$ $f'(3)=16\neq 0$ quadratic convergence.

~~linearly~~.

for $r=-1$

for $r=3$ $M=\frac{16}{32}=\frac{1}{2}$ $e_{i+1}\approx\frac{1}{2}e_i^2$

~~let: $x=\frac{x^3-x^2-3}{5}=g(x)$~~

$f''(x)=6x-2$ $f''(-1)=-8\neq 0$

~~$g'(x)=\frac{3x^2-2x}{5}$~~

~~$g(-1)=1$~~

~~not linearly.~~

$r=-1$ is a double root

$m=2$

$\lim\frac{e_{i+1}}{\cdot}<=\frac{m-1}{\cdot}-\frac{1}{\cdot}\Rightarrow e_{i+1}\approx\frac{1}{2}e_i$

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2. Proof: $f(x)=ax+b$ $\quad f'(x)=a$

initial point x_0

so $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = x_0 - \frac{ax_0+b}{a} = -\frac{b}{a}$

(independent from x_0)

And $f(x_1) = a(-\frac{b}{a})+b = 0$

3. (a) $x^3+x-2=0$, $x_0=0$

$f(x)=x^3+x-2$ $\quad f'(x)=3x^2+1$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0 - \frac{-2}{1} = 2$

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2 - \frac{8}{13} = \frac{18}{13}$

(b) $x^2 + \frac{1}{x+1} - 3x = 0$, $x_0 = 1$

$f(x) = x^2 + \frac{1}{x+1} - 3x$, $f'(x) = 2x - \frac{1}{(x+1)^2} - 3$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1 - \frac{-\frac{3}{2}}{-\frac{5}{4}} = -\frac{1}{5}$

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \approx 0.1809$

4. (a) $f(x)=x^2-a$ we know $f(\sqrt{a})=0$ $\quad (\sqrt{a}>0)$

Newton method:

$$x_{k+1} = x_k - \frac{x_k^2 - a}{2x_k}$$

$$= x_k - \frac{1}{2}x_k + \frac{1}{2}\frac{a}{x_k}$$

$$= \frac{1}{2}\left(x_k + \frac{a}{x_k}\right), \quad x_0 > 0$$

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(b) $\frac{x_{k+1}}{x_k} = \frac{1}{2}\left(1 + \frac{a}{x_k^2}\right)$ we know $x_k > \sqrt{a}$

so $\frac{a}{x_k^2} < 1$

so $\frac{x_{k+1}}{x_k} < 1 \Rightarrow x_{k+1} < x_k \Rightarrow \{x_k\}$ decreasing

5. Example : $f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} = 0$

(大概)

